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UDK: 007.52 **Research paper**

Comparison Between Different Humanoid Leg Hip Kinematics Trough Dynamic Simulation

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Abstract

A lot of existing humanoid robots have classical orthogonal leg joint kinematics where all pitch, roll and yaw joints are respectively along longitudinal, lateral and vertical axes of the robot's body. This paper investigates the influence of different rotary joint axis inclination at the hip joints on power distribution and torque versus angular speed for double support motions. The movements under study here are knees flexion/extension, swaying hips move and forward/backward pelvis motion. They are simulated using a dynamic simulator. This study can help to design better motor arrangement and therefore better power distribution.

Key words: *humanoid robotics, kinematics, dynamic simulation*

1. INTRODUCTION

Humans have more redundant degrees of freedom (DOF) than conventional humanoid robots. This enables them to achieve various motions with minimal energy. It is not possible to make an exact robotic copy of human kinematics. However the DOF configuration is one of the very important aspects for the design.
This research takes a place in the frame of the ROMEO project led by the French company Aldebaran-Robotics. One of the objectives of the project is to develop a humanoid robot that can act as a comprehensive assistant for persons suffering from loss of autonomy.
Existing human-size biped robots can achieve stable and dynamic walking. Many of them have classical leg joint kinematics where all pitch, roll and yaw joints are respectively along longitudinal, lateral and vertical axes of the robot's body. They have 3-DOF at the hip, 1-DOF at the knee and 2-DOF at the ankle. Examples of such humanoids are robots P2, P3, and ASIMO [1][2] from HONDA research group, HRP-2P [3][4] from the Japanese National Institute of Advanced Industrial Science and Technology (AIST) and Kawada Industries, Inc., H6 and H7 [5][6] from the University of Tokyo, WABIAN [7][8] from Waseda University, KHR-2 [9] from Korea Advanced Institute of Science and Technology (KAIST). HRP-2 has specific cantilever structure at the hip. WABIAN has an additional roll joint between hips. Figure 1 shows the classical hip joint orientation.

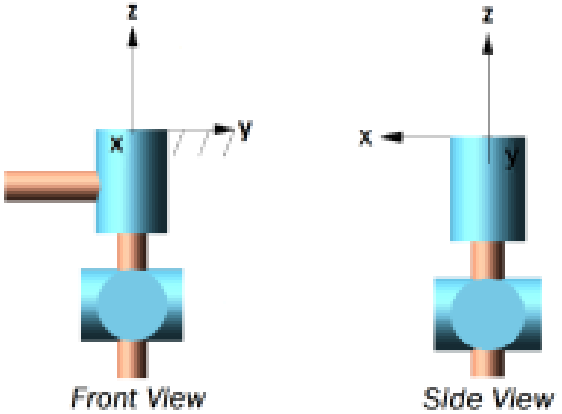


Figure 1. Classical hip joints orientation (front and side view).

All previously mentioned humanoids have classical joint axis orientation except Johnnie [10] from the Technical University of Munich and NAO [11] from the French Aldebaran Robotics company. The hip yaw joint axis of these two robots is inclined in the frontal plane (Fig. 2).

The hip yaw joint of Johnnie is inclined by 15° with respect to vertical. Hip yaw joints are coupled in the NAO robot and inclined by 45°.

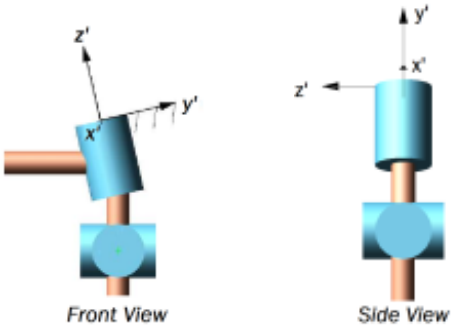


Figure 2. Inclination of hip yaw joint (front and side view).

This paper investigates the influence of different humanoid hip leg kinematics in matter of power consumption, maximal torque and torque versus speed for a variety of movements, namely knees flexion/extension, swaying hips move and forward/backward pelvis motion. The movements are simulated using a dynamic simulator. This study can help to design better motor arrangement and therefore better power distribution.
Section II deals with the robot's model that is used for the Romeo project and for simulation. Section III is dedicated to the description and setup of simulations. Results are described in section IV. Conclusion and perspectives are presented in section V.

2. HUMANOID ROBOT'S MODEL

2.1 Robot's model description

The robot is modelled and simulated in 3D with Adams software. It has 38 DoF, is 1.38 m tall and weighs 40.54 kg. Table [1] gives the main characteristics of the ROMEO humanoid. Each leg has 3 DOF at the hip, 1 DOF at the knee, 2 DOF at the ankle and 1 DOF at the toe. The order of joints from hip to foot is standard for humanoid robot, that is hip yaw, hip roll, hip pitch, knee pitch, ankle pitch, ankle roll and toe pitch. Each foot has two segments. The feet are equipped with 16 spheres that make contact with ground.

Table 1. Data of the robot's body parts.

Part	Masses *[Kg]*	Length *[m]*	DOF
Head	0.04 M_T	0.19 H_T	1
Arms (2)	0.26 M_T	0.4 H_T	14
Trunk	0.3 M_T	0.35 H_T	9
Legs (2)	0.4 M_T	0.46 H_T	14
Total	M_T =40.54	H_T = 1.38	38

Each sphere is associated with a contact sensor for measuring normal forces. Position of the sensors in the foot is presented in Fig. 3. This model of contacts enables to calculate the centre of pressure of the robot.

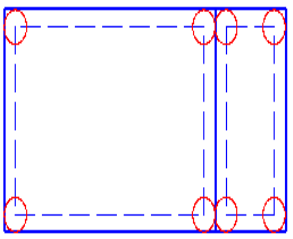


Figure 3. Position of sensors in the foot.

2.2 Model of contact

In Adams MSC.Software normal force F_n can be modelled as *restitution*, *impact* and *user defined.* The *restitution* model simulates a three-dimensional contact using coefficient of restitution for energy. For energy conservation *restitution coefficient* is set to 1. The *impact* model uses a three-dimensional contact with zero damping which can exhibit some dissipation of energy during the impact. The *user defined* lets the user define the force at contact by writing his own subroutine. The standard *impact* model was used here. The standard impact function is expressed as:

$$F_n = kp^e + c(p)\,dp/dt \quad (1)$$

where:

- p stands for the penetration of one geometry into another,
- k is stiffness,
- dp/dt is the penetration velocity at the contact point,
- e is a positive real value denoting the force exponent,
- $c(p)$ is damping as a function of penetration (see Fig.4) and
- p_{max} is a positive real value specifying the boundary penetration to apply the maximum damping c_{max} [12].

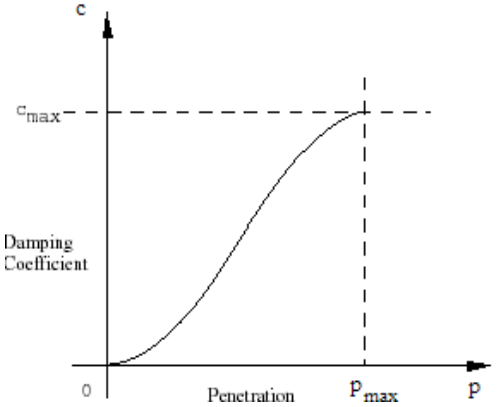


Figure 4.Damping coefficient versus penetration [12] (there is no citation number 14)

Table 2 shows the values of contact parameters for our model.

Table 2. Contact parameters used for simulation.

Parameter	Value
Stiffness	1.0E+010 [N/m]
Force Exponent	2.2
Damping c_{max}	8000 [Ns/m]
Penetration Depth p_{max}	1.0E-005 [m]
Static Coefficient μ_s	0.8
Dynamic Coefficient μ_d	0.76
Stiction Transition Velocity	0.1 [m/s]
Friction Transition Velocity	1 [m/s]

Static Coefficient of friction is μ_s = 0.8 and *Dynamic Coefficient of friction* is μ_d = 0.76. These two coefficients are related to the material type of the two contact bodies. When the slip velocity is equal to the value specified for *Friction Transition Velocity*, the coefficient of friction is the *Dynamic Coefficient*. Spheres under feet and ground are selected to be made of dry rubber.

2.3 Generation of model with different kinematics

This paper analyzes knees flexion/extension, swaying hips moves and forward/backward pelvis move. To save development time, Adams and Matlab/Simulink files are generated automatically. This enables to create models with modified kinematics very quickly. The starting file is a Python file from Aldebaran-Robotics Company that consists of all information that are necessary to build dynamic model such as shape of robot parts, weights, lengths, inertia, orientation of joints... First it is needed to execute Python file named GenerateMatlabFile LISV that outputs a Matlab file; this file contains the information of the initial Python file in Matlab syntax. Second, *generate Adams* must be executed to generate Adams ".cmd" file. The generated model of the humanoid robot is shown in Fig. 5.

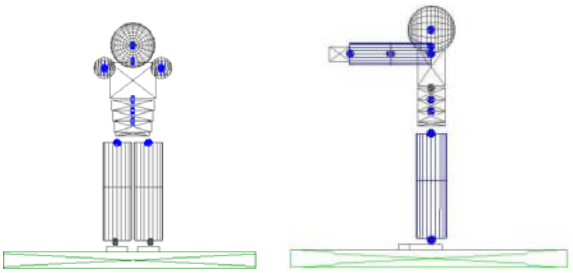


Figure 5. Wired model of the robot in Adams MSC.Software.

The generation of the ".mdl" file for Simulink/Matlab is done in a similar way as Adams ".cmd" file. Trajectories of desired angular position and desired angular velocity are calculated in Matlab by inverse kinematics and sent to Adams. The simulation in Adams gives output of the real angular position, angular velocity and torque from each joint that is moved, and normal forces from foot contact sensors below the feet.

3. SIMULATIONS

In classical kinematics of humanoid robot's legs all pitch joint axes are parallel and lateral. As a reference we use results from classical kinematics and compare them with results after inclination of hip joint axes. Configurations used for simulation are rotation about Z in clockwise direction by 15° and 30° and rotation about X by 15° and 30° in clockwise direction. Joint axes at the hip remain orthogonal (Fig. 6).

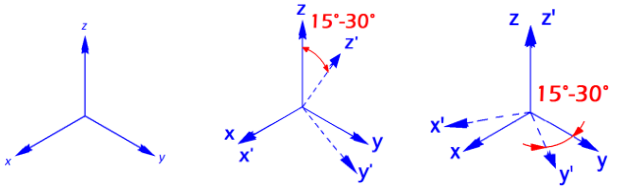


Figure 6. Different configurations at hip joints that are used for simulations.

All motors are controlled using a PID law. Dynamic simulation of the humanoid model gives us real information of leg's joint torques, angular position, angular velocity and normal forces in the feet that could be used for motor and gear selection.

3.1 First simulation: knees flexion/extension

The robot moves in the sagittal plane namely knee flexion and extension as shown in Fig. 7.

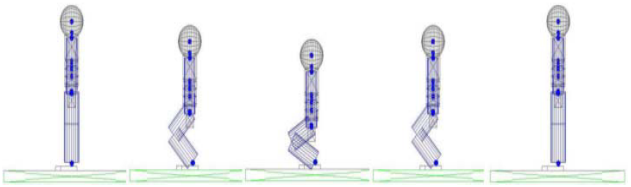


Figure 7. Simulation of knee flexion/extension.

The sequences of basic movements in the sagittal plane are the following:

1) Stay in initial position, 0.5 [sec]
2) Knees flexion on the spot, 1.5 [sec]
3) Stay in flexed position, 1 [sec]
4) Standing up from flexed position, 1.5 [sec]

The duration of each movement was set for the robot to achieve lively motion.

3.2. Second simulation: swaying hips

The robot moves, flexes knees and continues with the sequence of basic movements in the frontal plane (Fig. 8).

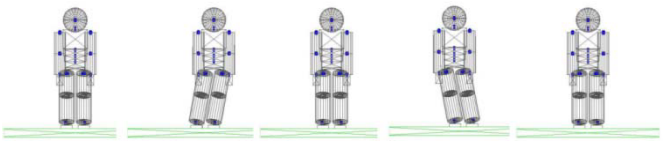


Figure 8. Simulation of sideways hip move.

The sequence of basic movements in the frontal plane is the following:

1) Move hip to the left side, 1.5 [sec]
2) Stay on left side 0.5 [sec]
3) Move hip to right side, 3 [sec]
4) Stay on right side, 0.5 [sec]
5) Go back to initial position, 1.5 [sec]

3.3. Third simulation: forward/backward pelvis move (Fig. 9)

The simulation of forward/backward pelvis motion is started when arms are down and knees are flexed.

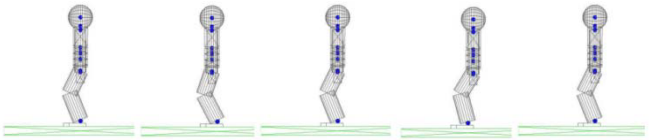


Figure 9. Simulation of forward/backward pelvis motion.

The sequence of basic movements for forward/backward pelvis move in the sagittal plane is the following:

1) Move pelvis forward, 1.5[sec]
2) Stay in forward position, 0.5[sec]
3) Move pelvis backward, 3[sec]
4) Stay in backward position, 0.5[sec]
5) Go back to start position, 1.5[sec]

We calculated power distribution during motions, thanks to information from the humanoid model. For each joint the data relative to all the simulated movements are grouped to get velocity variations as a function of torque over all the movements.
Power *P(t)* is related to the torque *T(t)* and angular velocity *W(t)* for each joint as:

$$P(t) = T(t) \cdot W(t) \qquad (2)$$

We use two criteria to compare results from simulation of different hip kinematics. The first criterion consists of considering torque versus speed in hip joints. The second criterion deals with power distribution.

4. SIMULATION RESULTS

The software used for simulation was Adams linked with Matlab/Simulink.

4.1. Knee flexion and extension - influence of the different rotary joint axes inclination at the hip rotary joints

The same simulation of knees flexion was carried out with classical hip kinematics orientation where all three joint axes are orthogonal, with rotation about Z of hip joint axis in clockwise direction by 15° and 30°, rotation about X in clockwise direction by 15° and 30°. Figures 10 - 13 present the results of power distribution at hip joints and torque versus speed for knee flexion and extension motion.

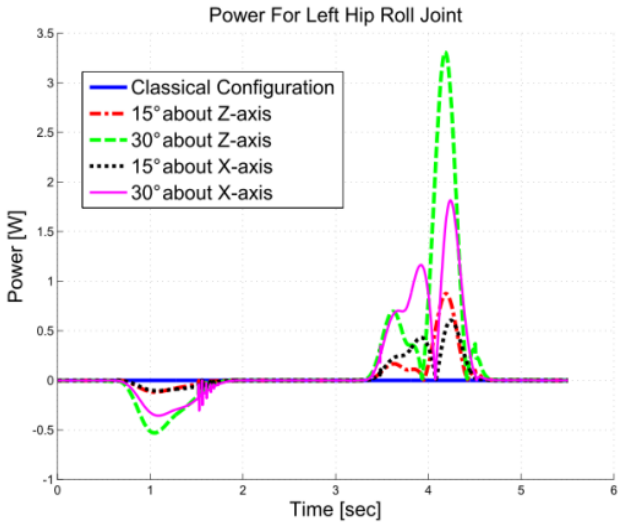


Figure 10. Power as a function of time for hip roll joint axis for the experiment of knee flexion and extension.

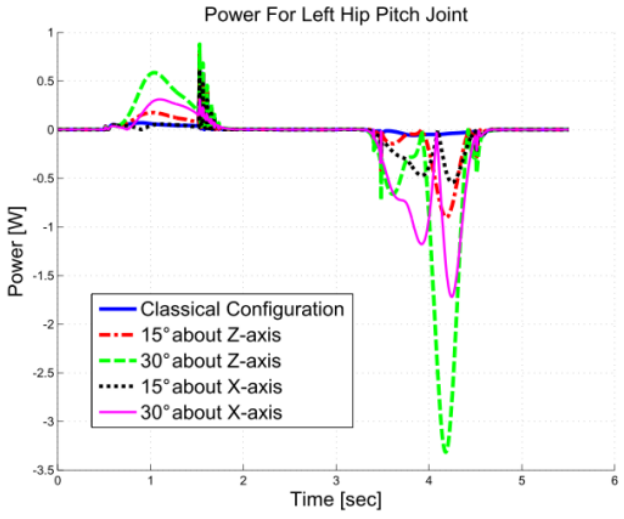


Figure 11. Power as a function of time for hip pitch joint axis for the experiment of knee flexion and extension.

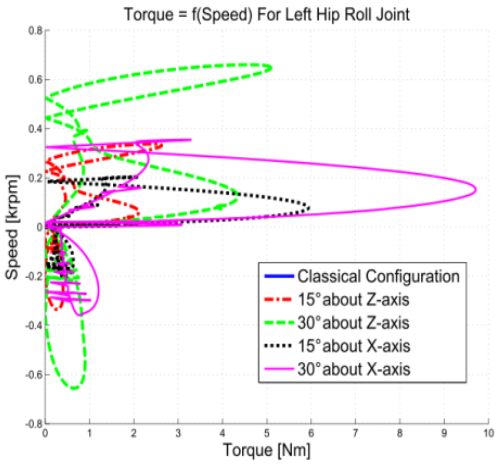


Figure 12. Torque versus speed for hip roll joint axis for the simulation of of knee flexion and extension.

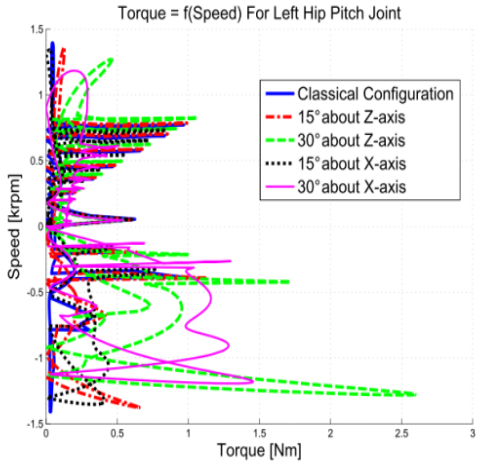


Figure 13. Torque versus speed for hip pitch joint axis for the simulation of knee flexion and extension.

The most affected hip joint axis in knee flexion and extension motion is the hip pitch. Both proposed kinematics for hip joint do not bring any benefit in power distribution. The humanoid robot needs 3 times more power at hip pitch and roll motors when the hip joint is rotated about Z axis by 30° and around 1.5 times when it is rotated about X axis by 30°.

The same configuration of hip joint when hip is rotated about X axis increases torque at hip roll joint axis until 10 Nm, compared with 0 [Nm] for the classical orientation of hip joint.

4.2 Swaying hips motion - influence of the different rotary joint axes inclination at the hip rotary joints

Figures 14 - 18 present the results of power distribution at hip joints and torque versus speed for swaying hips motion.

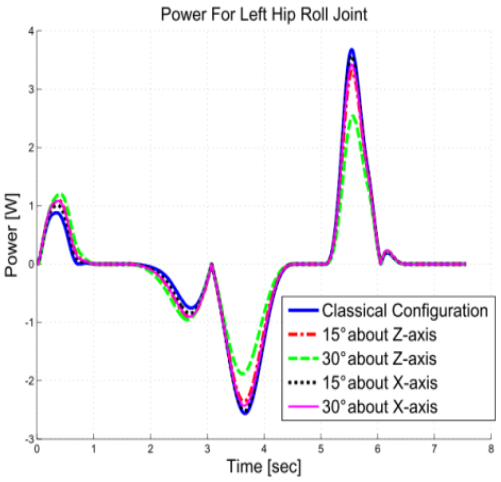


Figure 14. Power as a function of time for hip roll joint axis for the simulation of swaying hips motion.

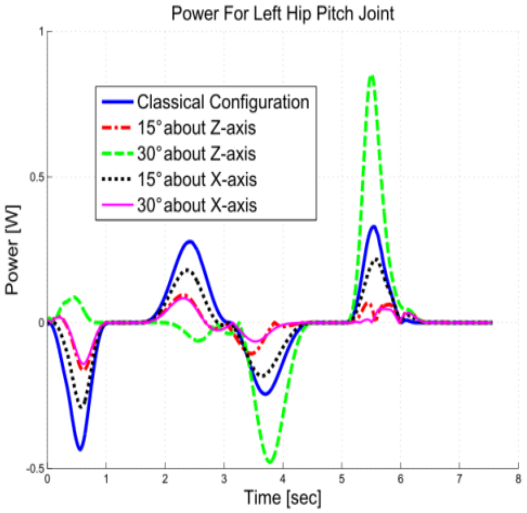


Figure 15. Power as a function of time for hip pitch joint axis for the simulation of swaying hips motion.

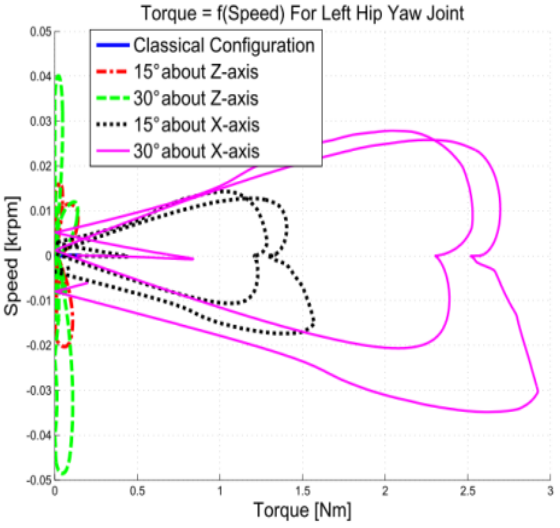


Figure 16. Power as a function of time for hip yaw joint axis for the simulation of swaying hips motion.

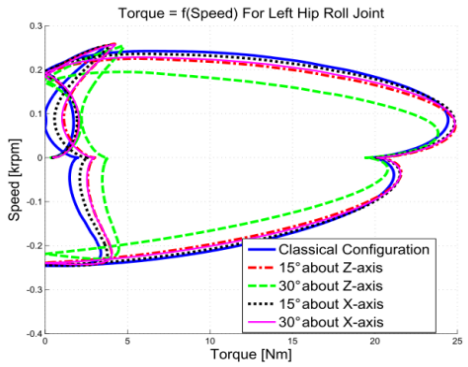


Figure 17. Torque versus speed for hip roll joint axis for the simulation of swaying hips motion.

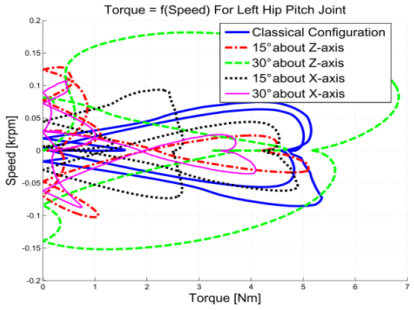


Figure 18. Torque versus speed for hip pitch joint axis for the simulation of swaying hips motion.

The most affected hip joint axis in swaying hips motion is the hip roll. The rotation of hip joint about X axis brings benefits just for power in hip pitch axis with a reduction of 20% for the 30 [deg] rotation. The rotation about Z decreases maximal torque at hip roll joint, but increases value of torque at hip pitch joint axis.

4.3 Forward/Backward pelvis motion-influence of the different rotary joint axes inclination at the hip rotary joints

Figures 19 - 21 present the results of power distribution at hip joints and torque versus speed for forward/backward pelvis motion.

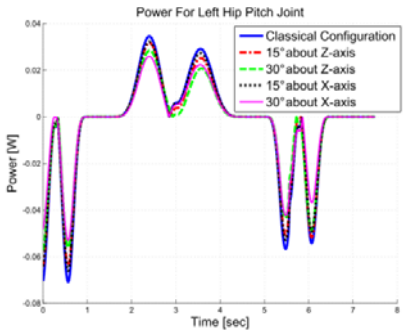


Figure 19. Power as a function of time for hip pitch joint axis for the simulation of forward/backward pelvis motion.

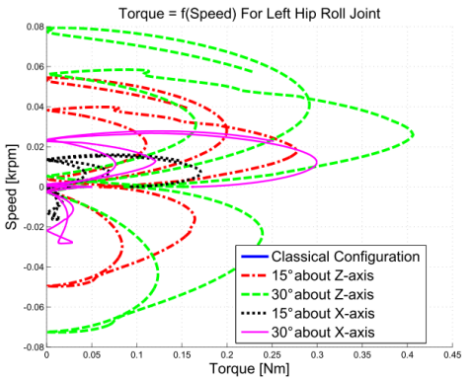


Figure 20. Torque versus speed for hip roll joint axis for simulation of forward/backward pelvis motion.

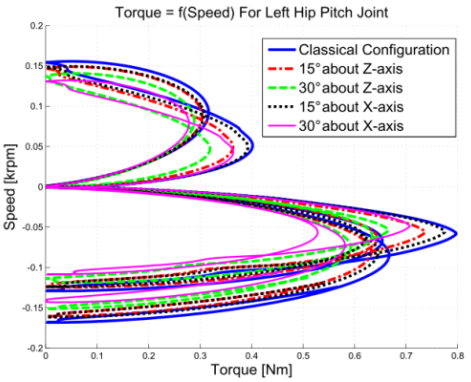


Figure 21. Torque versus speed for hip pitch joint axis for simulation of forward/backward pelvis motion.

Power and torque have very small values for the forward/backward pelvis motion, because this motion has a very small amplitude and different hip joint kinematics do not have big influence. Robot's trunk is a rigid body, so if the robot moves more backward it will fall down.

5. CONCLUSION

The main achievement of this work was to investigate the influence of different rotary joint axis inclination at the hip joints on power distribution and torque versus speed for knee flexion/extension, swaying hips move and forward/backward pelvis move. We investigated differences between classical hip joint axes configuration where all joint axes are orthogonal, rotations about Z by 15° and 30° in clockwise direction and rotation about X by 15° and 30° in clockwise direction.

There is no influence on power distribution and torque if the motion has small amplitude. Both proposed kinematics of hip joint do not bring any benefit for knee flexion and extension motion. When the angle of rotation of hip joint axes increases, torque and power generally increase. The rotation of hip joint about X axis gives better results for torque at hip pitch joint axis for

the swaying hips motion. Rotation of hip joint about Z, for the same motion, helps to reduce torque and power distribution for the hip roll joint axis, but increases them proportionally for the hip pitch joint axis.

Next developments will focus on the influence on power distribution and torque versus speed of motions when torso is active in such cases where the robot will have to sit down, grab object [13] from the ground and execute walking steps.

ACKNOWLEDGMENTS

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Poređenje različitih kinematika kuka humanoidnih robota putem dinamičke simulacije

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Primljen (05.06.2011.); Recenziran (26.10.2011.); Prihvaćen (20.10.2011.)

Rezime

Veliki broj postojećih humanoidnih robota ima klasičnu pravougaonu kinematiku nožnih zglobova gde se svi zglobovi nalaze duž uzdužnih, bočnih i vertikalnih osovina tela robota. Ovaj rad istražuje uticaj različitih rotacionih zglobnih nagiba osovina kod zglobova kuka na distribuciju snage i obrtni momenat naspram ugaone brzine za pokrete sa dvostrukim osloncem. Pokreti koji su deo ove studije su savijanje/istezanje kolena, njihanje kukova i napred/nazad pokret karlice. Svi su simulirani uz pomoć dinamičkog simulatora. Ova studija može da pomogne projektovanju boljeg raspoređivanja motora i time boljoj distribuciji snage.

Ključne reči: *humanoidna robotika, kinematika, dinamička simulacija*